Digital Economy and Industrial Employment Upgrading in China: Evidence from Provincial Panel Data, 2013–2022

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Abstract

This study asks whether China's digital-economy development is associated with industrial employment upgrading across 31 provinces, 2013–2022 ($N = 310$ province-years), using a principal-component digital-economy development index (DEDI) and two-way (province and year) fixed-effects regression. The central finding is compositional: digital development is robustly associated with upgrading within the urban-unit service sector toward higher-value advanced (information-technology, finance, scientific-research) services, surviving two-way fixed effects across all control sets and sample restrictions (the producer-service association is similar but weaker). By contrast, its link to the broad sectoral structure, which is characterized by a larger tertiary, smaller primary share, is a between-province, secular-trend pattern. This is strong cross-sectionally but differenced to null under fixed effects. A Hausman test selects fixed effects; a multilevel growth model (descriptive) shows heterogeneous provincial trajectories. Interpreted associationally, the digital economy appears linked less to gross sectoral reallocation than to upgrading within services—cautioning against reading the cross-sectional digital–tertiary correlation as a within-province effect.

Keywords: digital economy, employment structure, producer services, panel data, fixed effects

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Introduction

Since the early 2010s, China's digital economy (broadband and mobile internet, e-commerce, digital payments, and data-driven business models) has expanded rapidly and reorganized how firms produce and how labor is allocated across sectors. A large literature argues that digital technologies reshape employment through several channels: creating jobs on platforms and in digital services, substituting for routine tasks through automation, and reallocating labor across sectors, often with pronounced regional heterogeneity (Lu & Zhou, 2023; Wu & Yang, 2022; Xia & Pei, 2021; Ye et al., 2021; Zhao & Said, 2023). Understanding whether and how digitalization is connected to the structure of employment is central to debates about growth, inequality, and industrial policy in the world's second-largest economy.

This paper studies industrial *employment upgrading*—the movement of employment out of the primary sector and toward the tertiary sector, and, within services, toward higher-value producer services. The motivation is that employment upgrading is not only about workers' educational attainment but about *where* people work: whether labor moves out of agriculture into services, and whether the service sector itself becomes more producer-oriented and knowledge-intensive. A key conceptual distinction, often blurred in applied work, is between gross service expansion (retail, accommodation, public services) and producer-service upgrading (information technology, finance, scientific research, business services). Distinguishing these, and separating cross-province "between" variation from within-province "within" variation, is the paper's main contribution and remains under-examined for China at the provincial level. Anticipating the findings, digital development is robustly associated with upgrading *within* services toward high-value activities, whereas its association with the broad sectoral structure is a between-province pattern that fixed effects difference to null.

Research Questions

Primary RQ. How is provincial digital-economy development associated with industrial employment upgrading (measured chiefly by the tertiary employment share, with the primary and secondary shares as parallel outcomes) in China from 2013 to 2022?

Secondary RQ 1. Within the service sector, is digital development more strongly associated with *producer-service* employment (information technology, finance, scientific research, business services) than with consumer- or public-service employment?

Secondary RQ 2. Does the association differ between coastal and inland provinces?

Data and Variables

Source, Population, and Sample

The analysis uses a province-by-year panel compiled from National Bureau of Statistics of China (NBS) publications (the China Statistical, Labour Statistical, Tertiary Industry, Education, and Science and Technology Statistical Yearbooks), supplemented with the Peking University Digital Financial Inclusion Index (Guo et al., 2020) and provincial statistical communiqués (NBS, 2014–2023). The data represent the 31 provincial-level administrative units of mainland China (excluding Hong Kong, Macao, and Taiwan). The unit of analysis is the province-year; the sample spans 2013–2022, giving $31 \times 10 = 310$ observations.

Data Cleaning

The dependent-variable, control, and digital-economy workbooks were merged on province identifier and year. Variable names were standardized (e.g., *gov_intervention*, *investment_gdp_ratio*). Within-service shares were constructed on a consistent base of total classified urban-unit service employment; an advanced-service share and an industrial-upgrading index were derived (defined below). Nine province-years (Shaanxi, 2013–2018; Xinjiang, 2013–2015) have broad sector shares summing to less than 0.97 because reported sector employment falls short of total employment; these are flagged and retained, and the upgrading index uses sum-normalized shares. A pension/medical insurance ratio that exceeds one in 77 observations was excluded from the analysis. The final analytic panel has no missing values.

Variables

Outcomes. The broad industrial outcomes are the tertiary, primary, and secondary employment shares (sector employment $\div$ total provincial employment), summarized by an industrial-upgrading index, $1 \cdot (\text{primary share}) + 2 \cdot (\text{secondary share}) + 3 \cdot (\text{tertiary share})$, computed on shares rescaled to sum to one (range 1–3; higher $=$ more upgraded). For Secondary RQ 1, classified urban-unit service employment is partitioned into four mutually exclusive groups (producer, consumer, public, and mixed [real estate]), and each group's share is its employment $\div$ the classified service total; the producer-, consumer-, and public-service shares are the focal outcomes (the mixed share is retained only to complete the partition). The advanced-service share, a subset of the producer-service share, equals (information technology $+$ finance $+$ scientific research and technical services) urban-unit employment $\div$ the same denominator.

Predictor. The digital-economy development index (DEDI) summarizes 22 indicators organized under three dimensions—digital infrastructure, digital industry, and the digital environment (He et al., 2023). Heavily skewed indicators are log-transformed and all are z-standardized; principal component analysis (PCA) is applied to their correlation matrix (Kaiser–Meyer–Olkin $=$ .90; Bartlett's test of sphericity $p <$ .001), and four components meeting the Kaiser and 80%-variance criteria are retained, explaining 85.8% of the total variance. DEDI is the eigenvalue-share-weighted, sign-aligned sum of the four unrotated component scores, rescaled to $[0, 100]$. The full indicator system, equations, and PCA diagnostics are reported in Appendix A.

Controls. (Log) GDP per capita, urbanization rate, (log) total employment, trade openness, financial development, and government intervention. A coastal indicator (the 12 sea-bordering provinces) supports Secondary RQ 2. In all regressions, continuous predictors are standardized (mean 0, *SD* 1), so coefficients are interpreted per 1 *SD*; GDP per capita and total employment enter in logs. Because the DEDI *SD* is 17.8 index points, dividing a per-*SD* DEDI coefficient by 17.8 gives the coefficient per point on the original 0–100 scale (reported in the table notes).

Exploratory Data Analysis

Table 1 reports descriptive statistics. The mean tertiary employment share is 0.45 ($SD =$ 0.10); DEDI averages 48.7 ($SD = 17.8$) on its 0–100 scale. Table 2 reports pairwise correlations. DEDI is positively correlated cross-sectionally with the tertiary ($r = .34$), producer- ($r = .52$), and advanced-service ($r = .53$) shares and with development controls, and negatively with the primary share ($r = -.62$)—a pattern that motivates separating between- from within-province variation (developed below). Table 3 summarizes data completeness. Figure 1 shows that DEDI rose in every province yet the between-province gap stayed large and the provincial ordering stable, and Figure 2 plots every province's decade-long DEDI trajectory against the broad sector shares (Panel A) and the within-service shares (Panel B), making visible both a steep cross-province gradient and short within-province paths: descriptively, the tertiary, secondary, producer-, and advanced-service shares rise with DEDI while the primary share falls. These cross-sectional patterns mix between- and within-province variation and are disentangled by the models below.

Table 1

Descriptive Statistics for the Provincial Panel Data (2013–2022)

Variable	*M* (*SD*) / *n* (%)
Industrial employment structure	
Tertiary employment share	0.45 (0.10)
Primary employment share	0.29 (0.13)
Secondary employment share	0.25 (0.09)
Industrial upgrading index	2.15 (0.22)
Service-sector composition (urban units)	
Producer-service share	0.28 (0.07)
Consumer-service share	0.13 (0.05)
Public-service share	0.54 (0.12)
Advanced-service share	0.15 (0.04)
Digital economy	
Digital economy index (DEDI)	48.73 (17.80)
Economic controls	
GDP per capita (10,000 CNY)	6.38 (3.07)
Urbanization rate	0.60 (0.12)
Total employment (log)	7.51 (0.87)
Trade openness	0.25 (0.26)
Financial development	1.55 (0.44)
Government intervention	0.29 (0.20)
Coastal location	
Coastal	120 (38.7)
Inland	190 (61.3)

Note. $N = 310$. Continuous variables report *M* (*SD*); categorical variables report *n* (%). Coastal location is time-invariant, so frequencies are province-years (each province contributes 10). Percentages may not total 100 because of rounding. DEDI is the 0–100 digital-economy development index, a principal-component composite of 22 indicators (higher = more digital).

Table 2

Pairwise Pearson Correlations Among Analysis Variables

Variable	1	2	3	4	5	6	7	8	9	10	11
1. DEDI											
2. Tertiary share	.34										
3. Primary share	−.62	−.77									
4. Secondary share	.57	.04	−.65								
5. Producer share	.52	.74	−.77	.29							
6. Advanced share	.53	.74	−.72	.21	.94						
7. Log GDP p.c.	.65	.72	−.85	.46	.75	.72					
8. Urbanization	.57	.76	−.83	.40	.89	.82	.87				
9. Log employment	.73	−.27	−.11	.48	.07	.06	.10	.08			
10. Openness	.54	.64	−.79	.47	.81	.70	.70	.77	.14		
11. Finance	−.18	.48	−.27	−.13	.28	.32	.25	.27	−.65	.19	
12. Gov. intervention	−.67	−.08	.36	−.46	−.39	−.34	−.39	−.52	−.74	−.34	.50

Note. $N = 310$ province-years. DEDI is the 0–100 digital-economy development index, a principal-component composite of 22 indicators (higher = more digital). Pairwise Pearson correlations on the pooled (cross-sectional and over-time) data.

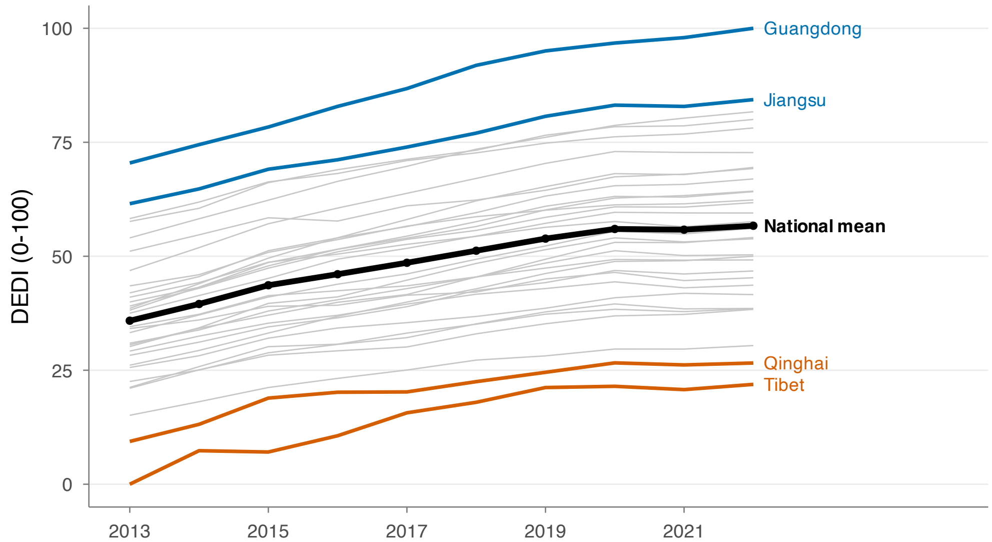
Table 3

Data Completeness and Source-Data Notes for the Analytic Panel

Item	Detail
Province-year observations	310 (31 provinces × 10 years), 2013–2022
Missing values on analysis variables	0 of 4,650 cells (0.0%)
Linearly interpolated source series	Some annual series in the source workbooks (documented in the cleaning log)
Sector shares summing < 0.97 (flagged)	9 province-years (Shaanxi 2013–18; Xinjiang 2013–15); retained, upgrading index uses sum-normalized shares
Variable excluded (measurement issue)	Insurance participation ratio (77/310 values > 1); not used

Figure 1

Digital-Economy Development Index (DEDI) by Province, 2013–2022



Note. Each thin gray line is one province's DEDI; the bold black line is the national (unweighted) mean. DEDI rose in every province over the decade, but the between-province gap remained large and the provincial ordering stable; the most-digital provinces (e.g., Guangdong, Jiangsu) and the least-digital (e.g., Tibet, Qinghai) stay far apart throughout, motivating the between-province focus of the analysis. DEDI is the 0–100 digital-economy development index, a principal-component composite of 22 indicators (higher = more digital).

Statistical Methods

Model 1: Fixed-Effects OLS

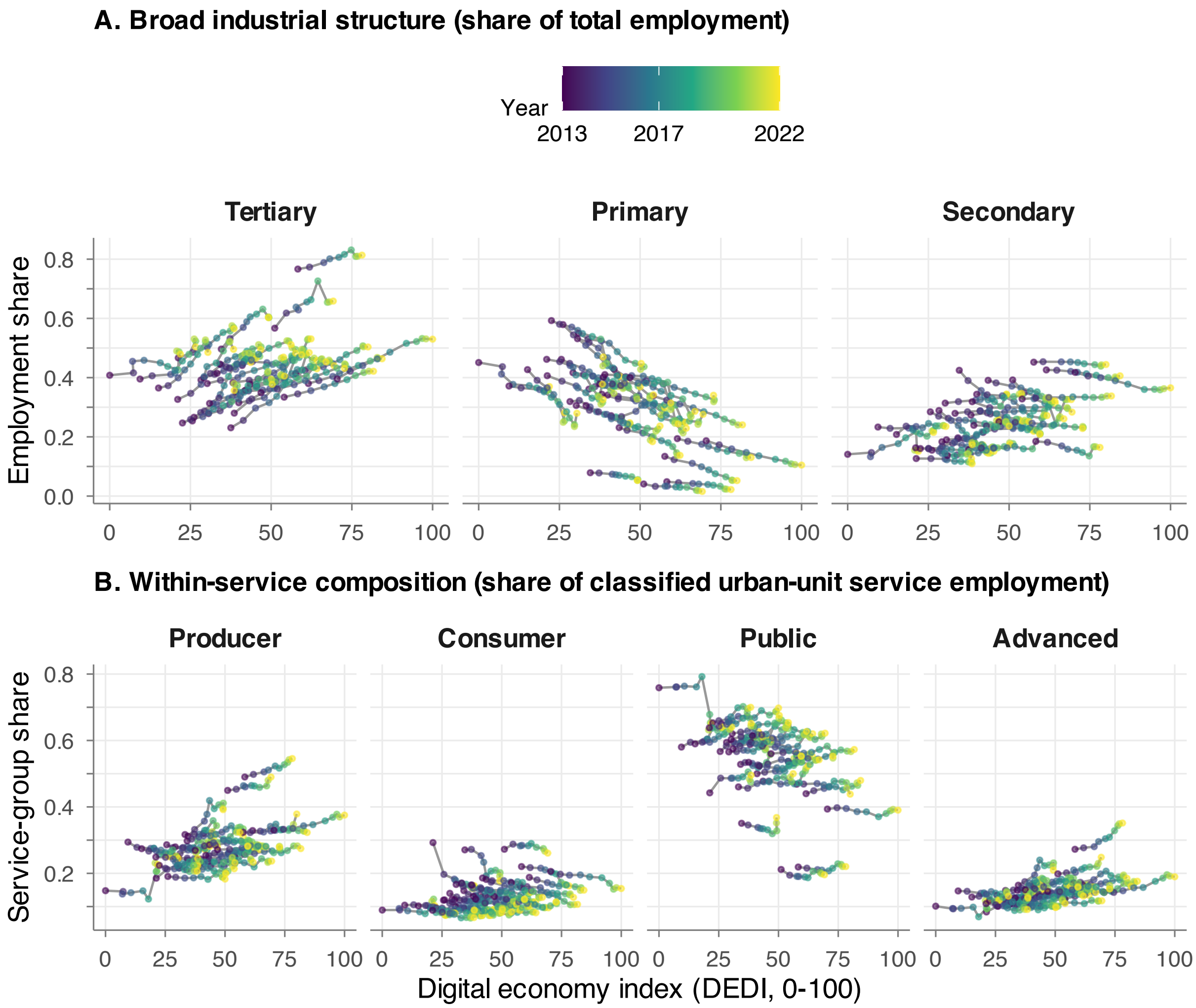
The primary model is an ordinary-least-squares (OLS) regression with two-way (province and year) fixed effects. For outcome Y_{it} in province i and year t,

$$Y_{it} = \beta_0 + \beta_1\,\mathrm{DEDI}_{it} + \boldsymbol{\theta}'\mathbf{X}_{it} + \alpha_i + \lambda_t + \varepsilon_{it}, \tag{1}$$

where $\mathbf{X}_{it}$ is the vector of standardized controls, α_i and λ_t are province and year fixed effects, and $\boldsymbol{\theta}$ collects control coefficients. The coefficient β_1 is the association between a 1-*SD* within-province change in digital development and the outcome, net of stable province characteristics and common national year shocks. Because DEDI is a slow-moving, largely between-province characteristic, this within estimate is conservative; the cross-sectional (between-province) association is therefore also reported descriptively, and a specification test (below) is used to choose between fixed and random

Figure 2

Province Trajectories of DEDI and Employment Structure: Broad Sectors (A) and Within-Service Composition (B)



Note. DEDI is the 0–100 digital-economy development index, a principal-component composite of 22 indicators (higher = more digital). Panel A: broad employment shares (of total employment). Panel B: within-service composition (of classified urban-unit service employment). Each connected path is one province's 10 annual observations across all 31 provinces; point color encodes year. The wide horizontal spread is between-province variation and each short path is within-province change, previewing the between-vs.-within distinction central to the results. Descriptively, the tertiary and secondary shares rise and the primary share falls with DEDI (Panel A), while the producer-, advanced-, and (more weakly) consumer-service shares rise and the public-service share falls (Panel B). These are raw cross-sectional patterns that mix between- and within-province variation; they are formally separated under two-way fixed effects in Tables 4–5 and Figure 3. The advanced-service share is a subset of the producer-service share.

effects.

The idiosyncratic error is assumed mean-independent of the regressors and fixed effects, $E[\varepsilon_{it} \mid \text{DEDI}_{it}, \mathbf{X}_{it}, \alpha_i, \lambda_t] = 0$. Rather than imposing independent, identically distributed homoskedastic errors, ε_{it} is allowed to be heteroskedastic and serially correlated within province but independent across provinces, and standard errors are therefore clustered by province. The other working assumptions are linearity in parameters, no perfect collinearity, and (for the within estimator) strict exogeneity, namely that the idiosyncratic error is uncorrelated with DEDI in all periods, not merely contemporaneously. Strict exogeneity is not directly testable and is the main threat to a causal reading; the analysis is consequently interpreted associationally. For Secondary RQ 2, a coastal interaction is added,

$$Y_{it} = \beta_0 + \beta_1 \,\text{DEDI}_{it} + \beta_2 \,(\text{DEDI}_{it} \times \text{Coastal}_i) + \boldsymbol{\theta}'\mathbf{X}_{it} + \alpha_i + \lambda_t + \varepsilon_{it}, \tag{2}$$

in which the time-invariant coastal main effect is absorbed by α_i, while β_2 captures how the within-province DEDI slope differs for coastal provinces.

Choosing Fixed Versus Random Effects

Because the random-effects (RE) estimator is consistent only if the province effects are uncorrelated with the regressors, the within and RE estimators are compared with a Hausman (1978) test and, because errors are clustered, a cluster-robust Mundlak (1978) correlated-random-effects test (a joint test of the within-province means of the regressors). A significant test favors fixed effects.

Model 2: Mixed-Effects (Multilevel) Growth Model

As an extension that uses the panel's nesting structure, a multilevel growth model is specified with province-specific random intercepts and random time slopes, with year centered at the first sample year ($t_c \equiv \text{Year}_t - 2013 \in \{0, \ldots, 9\}$):

$$Y_{it} = \gamma_0 + \gamma_1 \,\text{DEDI}_{it} + \gamma_2 \, t_c + \boldsymbol{\theta}'\mathbf{X}_{it} + u_{0i} + u_{1i} \, t_c + \varepsilon_{it}. \tag{3}$$

The random effects are assumed bivariate normal and the residual independent of them:

$$\begin{pmatrix} u_{0i} \\ u_{1i} \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \mathbf{G} \right), \qquad \mathbf{G} = \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{01} & \tau_1^2 \end{pmatrix}, \qquad \varepsilon_{it} \sim \mathcal{N}(0, \sigma^2). \tag{4}$$

The covariance $\mathbf{G}$ is left unstructured, so the random intercept and slope may correlate, $\rho_{01} = \tau_{01}/(\tau_0\tau_1)$ (because year is centered at 2013, τ_0 and ρ_{01} refer to the 2013 baseline, so $\rho_{01} < 0$ means provinces with a higher 2013 level upgrade more slowly). The level-1 residual is independent across observations and of the random effects (conditional independence given the random effects); a province-level AR(1) residual structure is a possible refinement that the present model does not impose. Estimation is by restricted maximum likelihood. Unlike Model 1, this model includes centered year as a linear term ($\gamma_2 t_c$) rather than year fixed effects, and it uses between-province as well as within-province variation to estimate γ_1; both features (the un-absorbed common trend and the between variation) make γ_1 less conservative than the two-way fixed-effects estimate, and comparing the two clarifies how much of the digital–structure association is within versus between provinces.

Results

The results are organized around *where* the digital–structure association lives. The headline, fixed-effects–robust result is compositional upgrading *within* the service sector (Secondary RQ 1); by contrast, the association with the *broad* sectoral structure (Primary RQ) is strong cross-sectionally but proves to be a between-province, secular-trend pattern that two-way fixed effects difference to null. The broad shares are reported first, their association is then localized to the between dimension, and the robust within-service estimates are presented last.

Primary RQ: Broad Industrial Shares

Table 4 reports the two-way fixed-effects estimates for the three broad shares. Under province and year fixed effects the DEDI coefficient is small and statistically insignificant for the tertiary ($\hat{\beta}_1 = -0.002$), primary (-0.022), and secondary (0.018) shares; urbanization is the dominant within-province correlate (tertiary +0.087, $p < .001$; primary −0.119, $p < .001$). This within-province null is not an absence of association but a relocation of it: as the next subsection shows, the strong digital–structure relationship is overwhelmingly between provinces and along the common national trend, and is differenced away once both fixed effects are included.

Table 4

Two-Way Fixed-Effects OLS: Digital Economy and Broad Industrial Employment Shares

	Tertiary	Primary	Secondary
DEDI (per *SD*)	−0.002 (0.025)	−0.022 (0.023)	0.018 (0.020)
Log GDP per capita	−0.018 (0.013)	−0.014 (0.010)	0.031 (0.009)**
Urbanization rate	0.087 (0.017)***	−0.119 (0.026)***	0.041 (0.019)*
Log employment	0.083 (0.050)	0.002 (0.050)	−0.035 (0.019)$^{+}$
Trade openness	−0.004 (0.012)	0.001 (0.011)	0.017 (0.010)
Financial development	−0.021 (0.006)**	0.003 (0.005)	0.016 (0.004)**
Government intervention	0.025 (0.011)*	−0.005 (0.016)	−0.014 (0.013)
Within R^2	0.20	0.38	0.39
Province FE / Year FE	Yes / Yes	Yes / Yes	Yes / Yes
N	310	310	310

Note. DEDI is the 0–100 digital-economy development index, a principal-component composite of 22 indicators (higher = more digital). Outcomes are employment shares (of total employment). On DEDI's original 0–100 scale (per-$SD \div 17.8$), the DEDI coefficients are −0.0001 (tertiary), −0.0012 (primary), and +0.0010 (secondary) per point. Within R^2 is the within-transformed (fixed-effects) R^2. Standardized predictors (per 1 *SD*); standard errors in parentheses; outcomes are proportions. $+\,p < .10$, $^{*}\,p < .05$, $^{**}\,p < .01$, $^{***}\,p < .001$.

Secondary RQ 1: Within-Service Upgrading

Table 5 shows a different picture for service composition. Holding province and year fixed, a 1-*SD* rise in DEDI is associated with a higher producer-service share (+0.029, $p < .10$) and, most clearly, a higher advanced-service share (+0.029, $p < .01$), a lower public-service share (−0.030, $p < .10$), and no change in the consumer-service share. Figure 3 displays these fixed-effects estimates and their within-province partial-regression counterparts. This selective upgrading toward higher-value advanced services is the paper's most robust finding (with producer services in the same direction but weaker).

Table 5

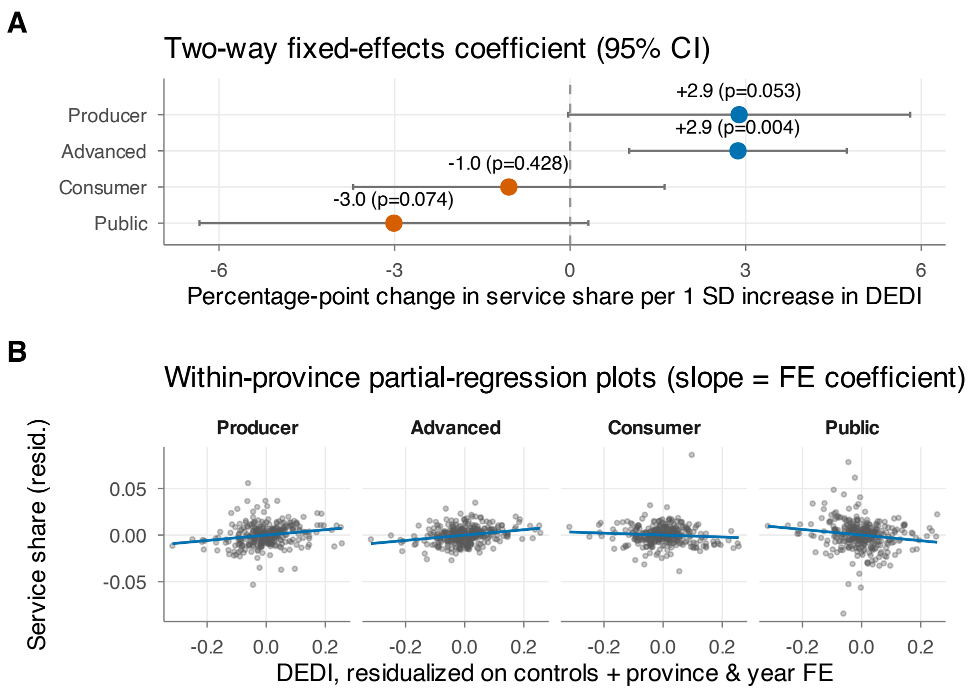
Two-Way Fixed-Effects OLS: Digital Economy and Within-Service Employment Shares

	Producer	Consumer	Public	Advanced
DEDI (per *SD*)	0.029 (0.014)$^{+}$	−0.010 (0.013)	−0.030 (0.016)$^{+}$	0.029 (0.009)**
Controls	Yes	Yes	Yes	Yes
Province FE / Year FE	Yes / Yes	Yes / Yes	Yes / Yes	Yes / Yes
Within R^2	0.23	0.09	0.16	0.29
N	310	310	310	310

Note. DEDI is the 0–100 digital-economy development index, a principal-component composite of 22 indicators (higher = more digital). Shares are of total classified urban-unit service employment, whose mutually exclusive groups are producer, consumer, public, and mixed (real estate; not shown); the advanced-service share is a subset of the producer-service share (its highest-value components: information technology, finance, and scientific research). Controls: (log) GDP per capita, urbanization, (log) total employment, trade openness, financial development, and government intervention. On DEDI's original 0–100 scale (per-*SD* ÷ 17.8), the DEDI coefficients are +0.0016 (producer), −0.0006 (consumer), −0.0017 (public), and +0.0016 (advanced) per point. Standardized predictors (per 1 *SD*); standard errors in parentheses; outcomes are proportions. $+\,p < .10$, $^{*}\,p < .05$, $^{**}\,p < .01$, $^{***}\,p < .001$.

Figure 3

Fixed-Effects Estimates for Within-Service Composition (Coefficients and Partial-Regression Plots)



Note. DEDI is the 0–100 digital-economy development index, a principal-component composite of 22 indicators (higher = more digital). Panel A: two-way fixed-effects coefficient (95% CI) per 1 *SD* of DEDI. Panel B: within-province partial-regression (added-variable) plots; each panel's slope equals the Panel-A coefficient. A single pooled regression line is intentionally avoided because it would conflate between- and within-province variation (Tufte, 2001). Panel A is in percentage points (per 1 *SD*); Panel B plots residual shares (1 unit = 100 pp), so the slope in each panel equals the same fixed-effects coefficient.

Why the Broad-Share Association Is Between, Not Within

The contrast between Tables 4 and 5 reflects where the identifying variation lies. DEDI is approximately 84% between provinces, and province and year dummies absorb about 99% of its variance, leaving only ∼1% within-province, off-trend variation. Table 6 traces the DEDI coefficient across estimators. For the tertiary share it is positive and significant under pooled OLS (0.086), random effects (0.050), and even one-way (province only) fixed effects (0.047), and collapses to zero only when year effects are added—i.e., the apparent effect is the common upward time trend, not idiosyncratic within-province variation. Producer- and advanced-service shares, by contrast, are positive under *both* the between and the two-way within estimators. Figure 4 visualizes this between-vs.-within comparison.

Table 6

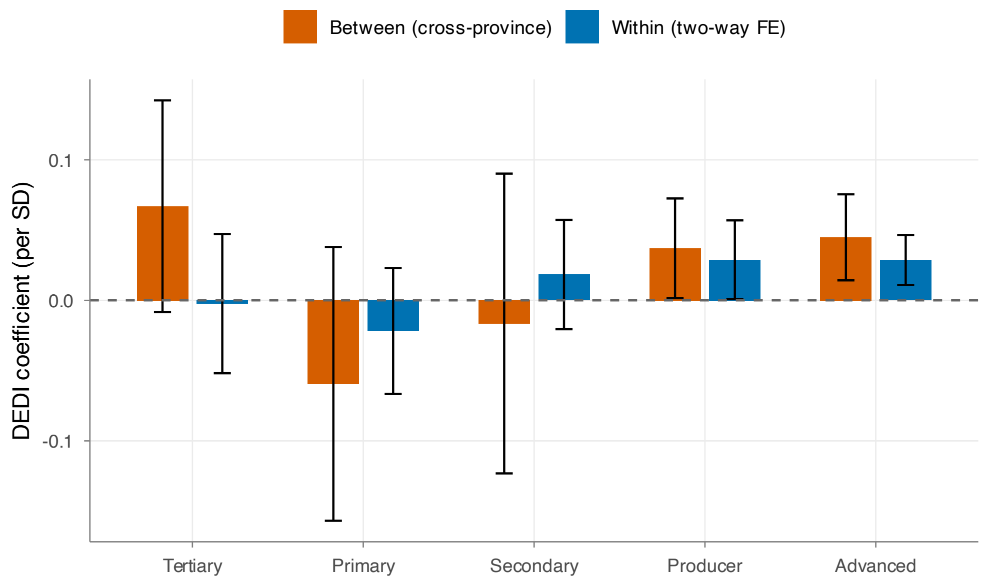
DEDI Coefficient (per SD) Across Estimators: Between Versus Within

Outcome	Pooled	Between	Random effects	Within (1-way)	Within (2-way)
Tertiary share	0.086**	0.067+	0.050*	0.047*	−0.002
Primary share	−0.044+	−0.059	−0.014	−0.008	−0.022
Secondary share	−0.043	−0.016	−0.030*	−0.035*	0.018
Producer share	0.006	0.037*	0.007	0.040***	0.029+
Advanced share	0.015	0.045**	0.020**	0.045***	0.029**

Note. DEDI is the 0–100 digital-economy development index, a principal-component composite of 22 indicators (higher = more digital). "Within (2-way)" is the preferred specification (province + year fixed effects). Standard errors are province-clustered for the pooled and within (fixed-effects) estimators and estimator-appropriate for the between (cross-province) and random-effects estimators. Coefficients are per 1 *SD* of DEDI; divide by 17.8 for the per-point (0–100) value. + $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$.

Figure 4

Between Versus Within DEDI Coefficient, by Outcome



Note. DEDI is the 0–100 digital-economy development index, a principal-component composite of 22 indicators (higher = more digital). Broad shares show a positive between-province but null within-province association; producer- and advanced-service shares are positive in both. Bars are coefficients per 1 *SD* of DEDI with 95% confidence intervals.

Specification Test: Fixed Versus Random Effects

Table 7 reports the fixed-versus-random-effects tests. Random effects are rejected for the tertiary ($p < .001$) and secondary ($p < .001$) shares; for the primary share the classical test does not reject and the robust Mundlak test is borderline ($p = .089$). Importantly, the rejection is driven by controls (labor force, openness, urbanization) correlated with province effects, not by DEDI itself: a DEDI-only Mundlak test does not reject for any outcome ($p \geq .48$)—it does not reject equality of DEDI's between- and within-province slopes, which is consistent with (though, given the thin within-province variation, not strong positive evidence for) treating the broad-share association as a between-province pattern rather than a within-province effect. Fixed effects are therefore the consistent choice; the positive RE/pooled DEDI–tertiary coefficient is a between-province artifact that fixed effects difference away. For the primary share, where random effects are not clearly rejected, fixed effects are retained for cross-equation comparability across the three broad shares; fixed effects remain consistent (if less efficient) when random effects are also valid.

Table 7

Fixed-Versus-Random-Effects Specification Tests

Outcome	Hausman $\chi^2(7)$, p	Mundlak $\chi^2(7)$, p	DEDI-only Mundlak p	Decision
Tertiary share	31.4, $< .001$	27.3, $< .001$	.63	Fixed effects
Primary share	6.2, .51	12.4, .089	.83	RE not rejected
Secondary share	75.0, $< .001$	52.4, $< .001$	.48	Fixed effects

Note. DEDI is the 0–100 digital-economy development index, a principal-component composite of 22 indicators (higher = more digital). Mundlak is a cluster-robust correlated-random-effects test (joint test of within-province means). The DEDI-only test isolates whether DEDI specifically is correlated with the province effects.

Secondary RQ 2: Coastal Versus Inland

Table 8 reports the coastal interaction. The implied within-province DEDI slopes are very similar for coastal and inland provinces, and the interaction is far from significant for both the tertiary ($p = .94$) and producer-service ($p = .83$) outcomes. Figure 5 shows the same: cross-sectionally the coastal gradient looks steeper, but that is a between-province feature; the within-province slopes essentially coincide. There is thus no evidence that the digital–services link is concentrated on the coast.

Table 8

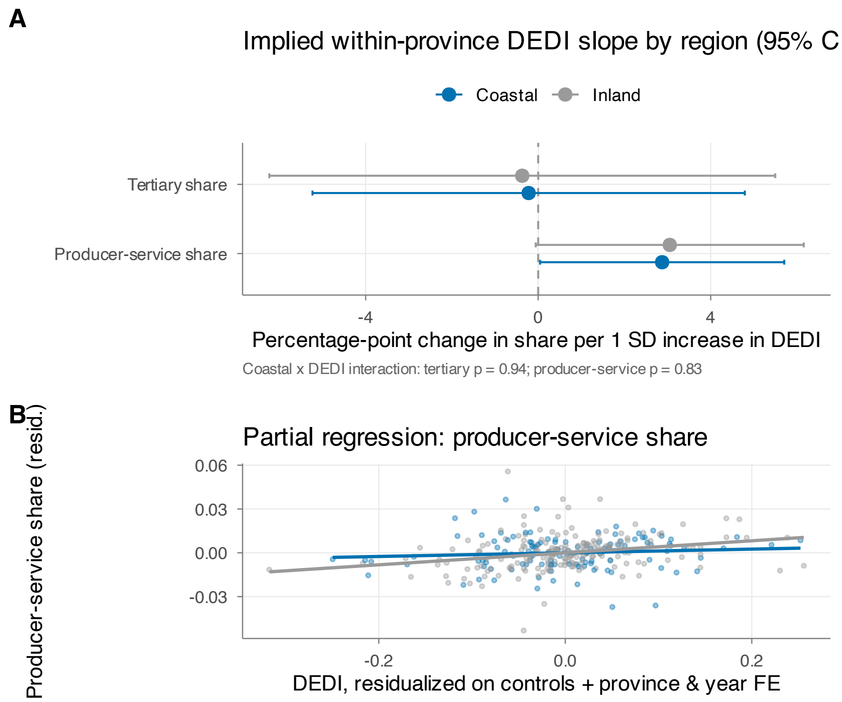
Coastal Versus Inland Heterogeneity (Two-Way Fixed Effects)

	Tertiary share	Producer-service share
Inland DEDI slope (β_1)	-0.004	0.031
Coastal DEDI slope ($\beta_1 + \beta_2$)	-0.002	0.029
Interaction (β_2: DEDI $\times$ Coastal), p	.94	.83

Note. DEDI is the 0–100 digital-economy development index, a principal-component composite of 22 indicators (higher = more digital). Slopes per 1 *SD* of DEDI (divide by 17.8 for the per-point, 0–100 value); the time-invariant coastal main effect is absorbed by province fixed effects. Coastal = 12 sea-bordering provinces.

Figure 5

Coastal Versus Inland: Fixed-Effects DEDI Slopes and Within-Province Partial Regression



Note. DEDI is the 0–100 digital-economy development index, a principal-component composite of 22 indicators (higher = more digital). Panel A: implied within-province DEDI slopes (95% CI) for inland and coastal provinces. Panel B: within-province partial regression for the producer-service share by group (near-parallel slopes).

Extension: Multilevel Growth Model

Table 9 compares the DEDI coefficient across the fixed-effects and multilevel estimators and reports the variance components. For the producer-service share the association is positive across all three estimators; for the tertiary share it is null under two-way fixed effects but positive under the multilevel models, which (like RE) also use between-province variation—consistent with the Hausman finding. The intraclass correlation is high (0.86 tertiary, 0.98 producer), confirming that most variation is between provinces. The random time slopes are non-singular and reveal heterogeneous provincial trajectories (Figure 6): the intercept–slope correlation is $-.71$ for the tertiary share, so provinces starting higher upgrade more slowly. Diagnostic checks indicate the estimated random effects are approximately normal, whereas the level-1 residuals are mildly non-normal; because this model is reported as a descriptive extension, the conclusions do not rest on its distributional assumptions.

Table 9

Multilevel Growth Models: DEDI Coefficient and Variance Components

Panel A. DEDI coefficient (per SD)

Outcome	Two-way FE OLS	ML random intercept	ML intercept + slope
Tertiary share	−0.002 (0.025)	0.021 (0.012)$^{+}$	0.041 (0.011)***
Producer-service share	0.029 (0.014)$^{+}$	0.036 (0.007)***	0.028 (0.007)***

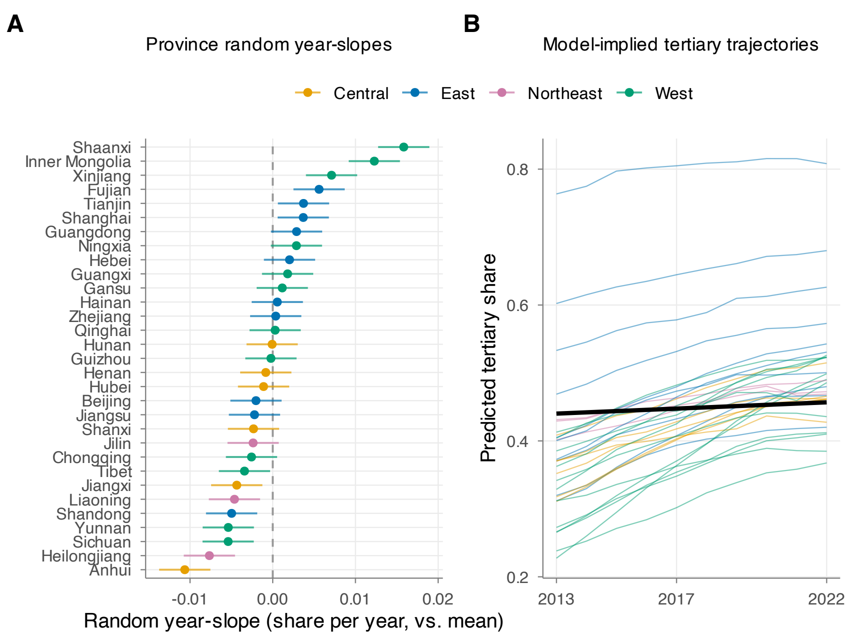
Panel B. Variance components (random intercept + slope model)

Outcome	τ_0 (intercept *SD*, 2013)	τ_1 (slope *SD*)	ρ_{01}	σ (residual *SD*)	ICC (RI model)
Tertiary share	0.064	0.006	−.71	0.015	.86
Producer-service share	0.029	0.005	−.21	0.010	.98

Note. DEDI is the 0–100 digital-economy development index, a principal-component composite of 22 indicators (higher = more digital). Estimated by restricted maximum likelihood (REML); both random-slope models are non-singular. The ICC column is the adjusted intraclass correlation from the random-intercept-only model; the τ_0, τ_1, ρ_{01}, and σ are from the random-intercept-plus-slope model, so the ICC does not equal $\tau_0^2/(\tau_0^2+\sigma^2)$ of the displayed values (a random-slope model has no single ICC, as it varies with year). Coefficients are per 1 *SD* of DEDI; divide by 17.8 for the per-point (0–100) value. Standardized predictors (per 1 *SD*); standard errors in parentheses; outcomes are proportions. $+\,p < .10$, $^{*}\,p < .05$, $^{**}\,p < .01$, $^{***}\,p < .001$.

Figure 6

Province-Specific Time Slopes (Caterpillar) and Model-Implied Tertiary-Share Trajectories



Note. Panel A: provinces ranked by their random year-slope (deviation from the average trend), with 95% confidence intervals. Panel B: model-implied tertiary-share trajectories (black line = average).

Robustness

Two checks support the headline pattern (Tables 10 and B1). First, the advanced-service association is significant under two-way fixed effects across the full sample and after dropping the four municipalities or the nine flagged low-sum province-years ($p \leq .007$); the producer-service association is positive but borderline ($p = .05$–.12), and the broad shares remain null throughout. Second, entering the four Varimax-rotated principal components underlying DEDI jointly, in place of the composite, leaves these conclusions unchanged and localizes the signal: the within-province advanced- and producer-service associations load on the industry/innovation component (RC1) and, for advanced services, the per-capita-penetration component (RC2), whereas the infrastructure-connectivity and enterprise-digitalization components are jointly insignificant and the broad tertiary share remains jointly null (Appendix B, Table B1).

Table 10

Sample Robustness of the Two-Way Fixed-Effects DEDI Coefficient (per SD)

Outcome	Full ($N = 310$)	Drop municipalities ($N = 270$)	Drop low-sum ($N = 301$)
Tertiary share	−0.002 (.93)	0.010 (.70)	0.005 (.84)
Primary share	−0.022 (.35)	−0.026 (.24)	−0.024 (.33)
Producer share	0.029 (.05)	0.028 (.07)	0.021 (.12)
Advanced share	0.029 (.004)	0.029 (.004)	0.024 (.007)

Note. DEDI is the 0–100 digital-economy development index, a principal-component composite of 22 indicators (higher = more digital). Each cell is the two-way fixed-effects DEDI coefficient (per 1 *SD*; divide by 17.8 for the per-point, 0–100 value) with its p-value in parentheses.

Discussion and Conclusion

This study asked whether China's digital-economy development is associated with industrial employment upgrading from 2013 to 2022, and the clearest result is compositional. Even under two-way fixed effects, which strip away fixed province characteristics and the common national trend, digital development is robustly associated with a shift *within* the service sector toward higher-value advanced services (and, more weakly, producer services), holding across every control set and sample restriction. In the part of the economy where digitalization plausibly acts most directly (the composition of service work), the data show a stable, identifiable signal of upgrading toward information-technology, finance, and scientific-research employment.

Several non-exclusive mechanisms could give rise to this within-province pattern. Digital infrastructure and platforms directly raise both the demand for and the supply of information-technology, data, and software services, which are themselves high-value service activities; digitalization also lowers the transaction and coordination costs of outsourcing finance, research, and business functions, plausibly deepening the producer-service division of labor (Wu & Yang, 2022; Ye et al., 2021). Because digital technologies tend to complement high-skill labor, digitalizing provinces may shift service employment toward knowledge-intensive activities while routine consumer- and public-service tasks grow more slowly (Lu & Zhou, 2023; Zhao & Said, 2023). These readings are interpretive rather than causal, and one caveat is structural: digital development and advanced services overlap conceptually, with DEDI partly capturing the same information-technology and digital-finance activity that the advanced-service share measures. The association is therefore best understood as a robust descriptive regularity of the knowledge economy rather than as an effect of digital infrastructure on service composition.

The *broad* sectoral structure tells an instructive second story. The strong cross-sectional association between digital development and a larger tertiary (and smaller primary) share is real but is a between-province, long-run pattern: it is positive across pooled, random-effects, and one-way fixed-effects estimators yet vanishes under two-way fixed effects, and the data cannot statistically separate DEDI's between and within slopes. This is not a non-result; it corrects a conflation common in applied work (reading a large cross-sectional digital–tertiary correlation as a within-province, causal-leaning effect) and shows that, net of what makes provinces persistently different and of the secular shift toward services, digitalization does not by itself move labor across the broad sectors in this decade.

Two further results round out the picture. There is no meaningful coastal–inland difference in the within-province association (Secondary RQ 2), so the digital–services link is not a coastal phenomenon; and the multilevel growth model, reported descriptively, reinforces the between-province interpretation (most variation in the shares lies between provinces, ICC $\geq .86$) while documenting heterogeneous upgrading trajectories across provinces.

Limitations

The design is observational, and the estimates are associational: fixed effects remove time-invariant province confounders but not time-varying confounders or reverse causality, and the within-province DEDI signal is thin and collinear with within-province trends in urbanization and income. The service-composition outcomes are based on urban-unit employees and so describe formal urban employment rather than the entire labor market. Some source series are interpolated, nine province-years carry a flagged accounting discrepancy, and the panel is modest ($N = 310$).

Future Work

Credible identification would require plausibly exogenous variation in digital development—for example, staggered infrastructure or policy rollouts usable as instruments or in difference-in-differences designs. Adding firm- or worker-level data and modeling province-level AR(1) residuals in the multilevel framework are natural extensions.

Conclusion

The digital economy in China appears connected less to gross sectoral reallocation than to *compositional upgrading within services*, toward higher-value advanced (and, more weakly, producer) activities. Separating between- from within-province variation is essential: it reconciles the large cross-sectional correlations with the muted within-province estimates and focuses attention on the result that survives two-way fixed effects—upgrading toward advanced services.

References

American Psychological Association. (2020). *Publication manual of the American Psychological Association* (7th ed.). American Psychological Association.

Guo, F., Wang, J., Wang, F., Kong, T., Zhang, X., & Cheng, Z. (2020). Measuring China's digital financial inclusion: Index compilation and spatial characteristics. *China Economic Quarterly, 19*(4), 1401–1418.

Hausman, J. A. (1978). Specification tests in econometrics. *Econometrica, 46*(6), 1251–1271.

He, D., Zhao, X., & Qi, Q. (2023). Measurement, spatial pattern and regional difference of digital economy development in China. *Industrial Technology & Economy, 42*(3), 54–62. [In Chinese.]

Kaiser, H. F. (1974). An index of factorial simplicity. *Psychometrika, 39*(1), 31–36.

Lu, Y., & Zhou, L. L. (2023). Can the digital economy improve the employment structure? A spatial Durbin model analysis. *Sustainability, 15*.

Mundlak, Y. (1978). On the pooling of time series and cross section data. *Econometrica, 46*(1), 69–85.

National Bureau of Statistics of China. (2014–2023). *China statistical yearbook* (and China labour, tertiary industry, education, and science & technology statistical yearbooks). China Statistics Press.

Tufte, E. R. (2001). *The visual display of quantitative information* (2nd ed.). Graphics Press.

Wu, B., & Yang, W. (2022). Empirical test of the impact of the digital economy on China's employment structure. *Finance Research Letters, 49*.

Xia, T., & Pei, J. (2021). The impact of digital economy on employment—Thinking based on the epidemic situation in 2020. *E3S Web of Conferences, 235*, 03034.

Ye, X., Du, Y., & He, W. (2021). The employment-structure effect of digital-economy development. *Finance and Trade Research, 32*(4), 1–13.

Zhao, Y., & Said, R. (2023). The effect of the digital economy on the employment structure in China. *Economies, 11*(9), 227.

Appendix A

Construction of the Digital-Economy Development Index (DEDI)

Indicator system and pre-processing. DEDI is built from 22 third-level indicators, organized under three first-level dimensions following He et al. (2023): digital infrastructure (6 indicators), digital industry (5), and the digital environment (11). Table A1 lists them. To limit the influence of a few provincial outliers (Beijing, Shanghai, Guangdong), the 12 indicators with absolute skewness above 1.5 are log-transformed, and all 22 are then z-standardized, giving a 310×22 matrix $\mathbf{Z}$ with zero-mean, unit-variance columns. (A small number of source cells were linearly interpolated over time before standardization, chiefly x_5, IPv4 addresses, and x_{13}, mobile internet users.)

Principal component analysis. Let $\mathbf{R}$ be the correlation matrix of $\mathbf{Z}$. PCA solves the eigenvalue problem

$$\mathbf{R}\,\mathbf{a}_j = \lambda_j\,\mathbf{a}_j, \qquad \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_{22} \geq 0, \tag{A1}$$

and forms component scores

$$\mathbf{F} = \mathbf{Z}\mathbf{A}\,(\mathbf{A}^{\top}\mathbf{A})^{-1}, \tag{A2}$$

where $\mathbf{A}$ collects the loadings of the k retained components. Sampling adequacy is high: the Kaiser–Meyer–Olkin measure (Kaiser, 1974) is .90, and Bartlett's test of sphericity rejects the identity hypothesis, $\chi^2(231) = 11{,}553.82$, $p < .001$. The number retained follows $k = \max(k_{\text{Kaiser}}, k_{80\%}) = 4$: four eigenvalues exceed one and four components explain 85.8% of the variance (Table A2; Figure A1). A Varimax rotation gives a simple-structure solution interpretable as (RC1) industry/innovation scale, (RC2) per-capita penetration, (RC3) infrastructure connectivity, and (RC4) enterprise digitalization intensity (Figure A2).

Composite index. The Varimax rotation above is used only to interpret and label the loadings; the index itself is built from the unrotated component scores. DEDI is the eigenvalue-share-weighted, sign-aligned sum of the four retained unrotated principal-component scores $F_{j,it}$, rescaled to $[0, 100]$:

$$\text{DEDI}_{it} = 100 \cdot \frac{\widetilde{F}_{it} - \min \widetilde{F}}{\max \widetilde{F} - \min \widetilde{F}}, \qquad \widetilde{F}_{it} = \sum_{j=1}^{4} w_j\, s_j\, F_{j,it}, \qquad w_j = \frac{\lambda_j}{\sum_{l=1}^{4} \lambda_l}, \tag{A3}$$

with eigenvalue-share weights $w = (.687, .154, .097, .062)$ and sign-flip vector $s = (+1, -1, +1, +1)$, so that higher values mean more digital for every component. In the 2019 cross-section the index ranges from Guangdong (95.0) and Jiangsu (80.7) at the top to Tibet (21.2) and Qinghai (24.5) at the bottom; Figure A3 shows the full province-by-year pattern.

Table A1

Hierarchical Indicator System for DEDI (22 Third-Level Indicators)

Dimension	Third-level indicators
Digital infrastructure (x_1–x_6)	Optical cable; broadband ports; mobile base stations; domain names; IPv4 addresses; websites
Digital industry (x_7–x_{11})	Software revenue; telecom revenue; websites per 100 enterprises; % e-commerce firms; e-commerce sales
Digital environment (x_{12}–x_{22})	Computers per 100 people; mobile internet users; mobile phone subscribers; digital TV; digital inclusive finance; digital–real integration; % IT employment; bachelor's graduates; R&D personnel; R&D institutions; patent grants

Note. These are the 22 indicators from which DEDI is built; hierarchy follows He et al. (2023). x_j denotes the jth indicator.

Table A2

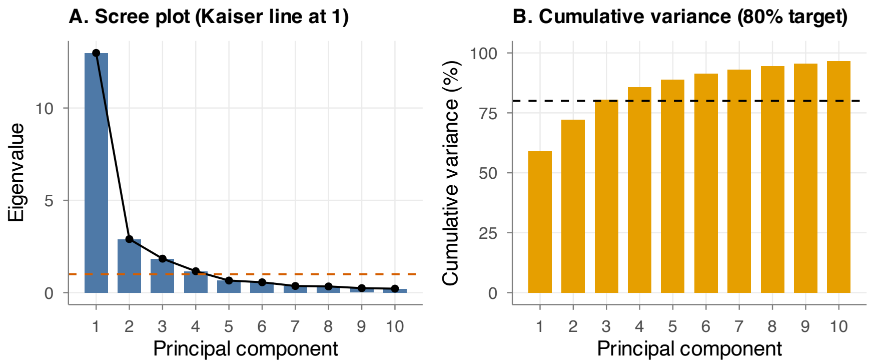
Eigenvalues and Variance Explained (First Five Principal Components)

PC	Eigenvalue	Variance (%)	Cumulative (%)
1	12.98	59.0	59.0
2	2.90	13.2	72.2
3	1.84	8.4	80.6
4	1.16	5.3	85.8
5	0.66	3.0	88.8

Note. From the principal-component analysis of the 22 indicators underlying DEDI (Table A1). Four components (eigenvalue ≥ 1 and cumulative variance $\geq 80\%$) are retained.

Figure A1

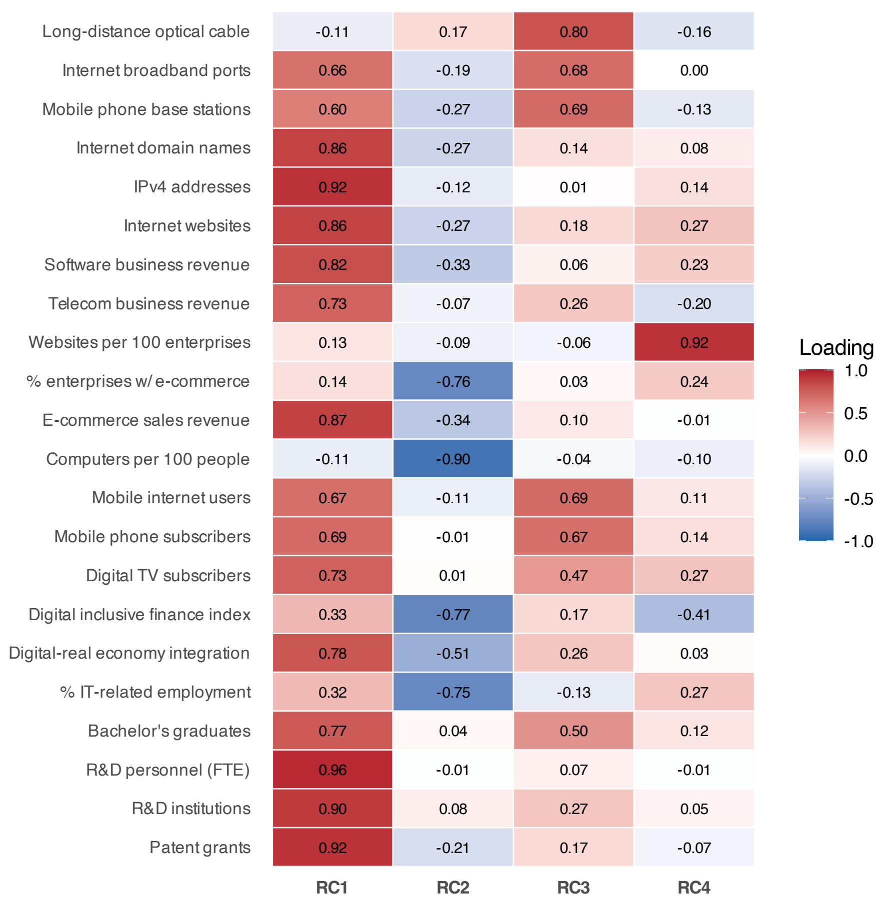
PCA Retention Diagnostics: Scree Plot and Cumulative Variance



Note. From the PCA of the 22 indicators underlying DEDI. Left: scree plot (dashed line at eigenvalue = 1). Right: cumulative variance explained (dashed line at the 80% target). Four components are retained.

Figure A2

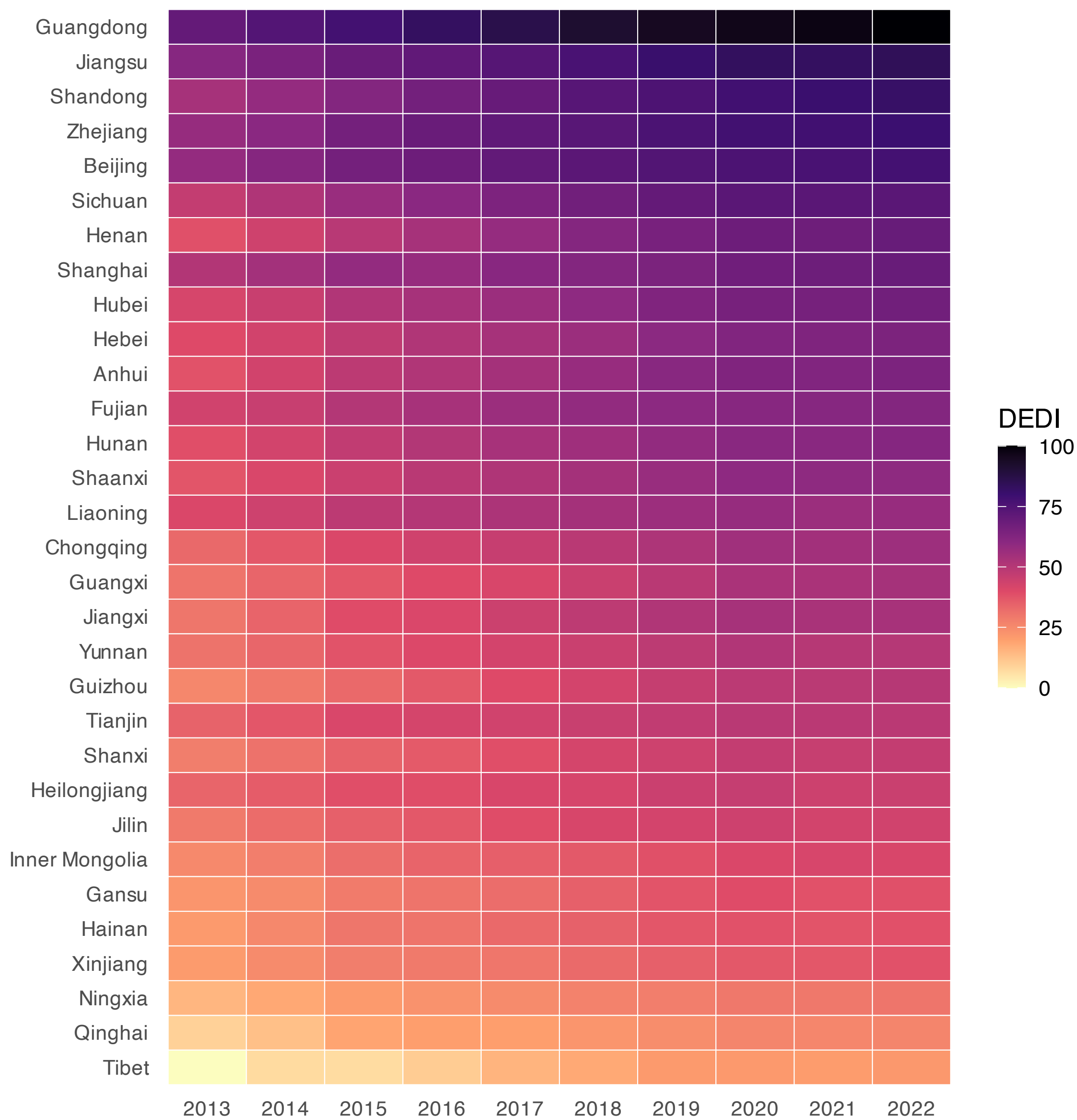
Varimax-Rotated Loadings of the 22 Indicators on the Four Retained Components



Note. Cell shading is each indicator's Varimax-rotated loading (its correlation with the component) in the PCA underlying DEDI. RC1, industry/innovation scale; RC2, per-capita penetration; RC3, infrastructure connectivity; RC4, enterprise digitalization intensity.

Figure A3

DEDI by Province and Year (Provinces Ordered by 2022 DEDI)



Note. DEDI is the 0–100 digital-economy development index, a principal-component composite of 22 indicators (higher = more digital). Each cell is a province-year DEDI value (0–100; magma color scale, light = low, dark = high). Provinces are ordered by their 2022 DEDI. The stable top-to-bottom ordering (Guangdong, Jiangsu, Shandong . . . Qinghai, Tibet) reflects the persistent east–west between-province gradient, while the uniform left-to-right darkening shows that every province rose over the decade—the between-province dominance that motivates the two-way fixed-effects design.

Appendix B

Decomposing DEDI: The Four Rotated Components as Separate Predictors

As a check on whether the within-province service-upgrading association reflects a particular facet of digitalization rather than the composite, the two-way fixed-effects models are re-estimated, entering the four Varimax-rotated principal components underlying DEDI (RC1, industry/innovation; RC2, per-capita penetration; RC3, infrastructure connectivity; and RC4, enterprise digitalization; see Appendix A) jointly, each standardized to unit variance, in place of the single index. DEDI itself is an eigenvalue-share-weighted, sign-aligned combination of the four *unrotated* principal components (Equation A3); the Varimax rotation spans the same four-dimensional component subspace, so entering the rotated components jointly tests exactly the same subspace as the index while giving an interpretable (simple-structure) basis. Because the rotation is orthogonal, the joint Wald test below is identical whether the rotated or the unrotated components are entered; the rotated basis is used here only because its components are individually interpretable. Controls, fixed effects, and province-clustered standard errors are as in Tables 4–5. Table B1 reports the results for the three focal outcomes.

The conclusions are unchanged, and the decomposition is informative. For the broad tertiary share, the four components are jointly insignificant under two-way fixed effects (Wald $p = .95$), echoing the null in Table 4. For the within-service outcomes, the association loads on the *software and application* side of digitalization rather than the *physical-infrastructure* side: the advanced-service share rises with both the industry/innovation component (RC1, +0.028 per *SD*, $p = .010$) and the per-capita-penetration component (RC2, +0.018, $p = .023$), and the producer-service share rises with the industry/innovation component (RC1, +0.056, $p < .001$; the infrastructure component RC3 is only borderline, $p = .059$). The infrastructure-connectivity and enterprise-digitalization components carry no robust within-province signal. The producer-service association, borderline with the composite (Table 5, $p = .05$), is sharp on RC1, because the composite spreads the industry/innovation signal across all four sign-aligned dimensions, whereas RC1 isolates it. In short, the within-province link to high-value service upgrading runs through the digital-industry/innovation and digital-application dimensions, consistent with the complementarity and skill-bias mechanisms in the Discussion.

Table B1

Two-Way Fixed-Effects Decomposition: The Four Rotated Components as Joint Predictors (per SD)

Rotated component	Tertiary	Advanced	Producer
RC1 (industry/innovation)	−0.006 (0.031)	0.028 (0.010)*	0.056 (0.015)***
RC2 (per-capita penetration)	0.010 (0.016)	0.018 (0.007)*	0.012 (0.009)
RC3 (infrastructure connectivity)	−0.005 (0.012)	0.005 (0.005)	0.012 (0.006)$^{+}$
RC4 (enterprise digitalization)	−0.004 (0.007)	0.006 (0.005)	0.005 (0.005)
Joint Wald (p, 4 RCs)	.95	.014	.009
Controls / Province FE / Year FE	Yes / Yes / Yes	Yes / Yes / Yes	Yes / Yes / Yes
N	310	310	310

Note. DEDI is the 0–100 digital-economy development index, a principal-component composite of 22 indicators (higher = more digital). The four Varimax-rotated components (Appendix A) are entered jointly, each standardized to unit variance. Tertiary is the broad tertiary share (of total employment); advanced and producer are shares of classified urban-unit service employment. RC2 is sign-oriented so that higher values denote greater per-capita digital penetration. Standardized predictors (per 1 *SD*); standard errors in parentheses; outcomes are proportions. $+\,p < .10$, $^{*}\,p < .05$, $^{**}\,p < .01$, $^{***}\,p < .001$.